Name:

Matriculation number:

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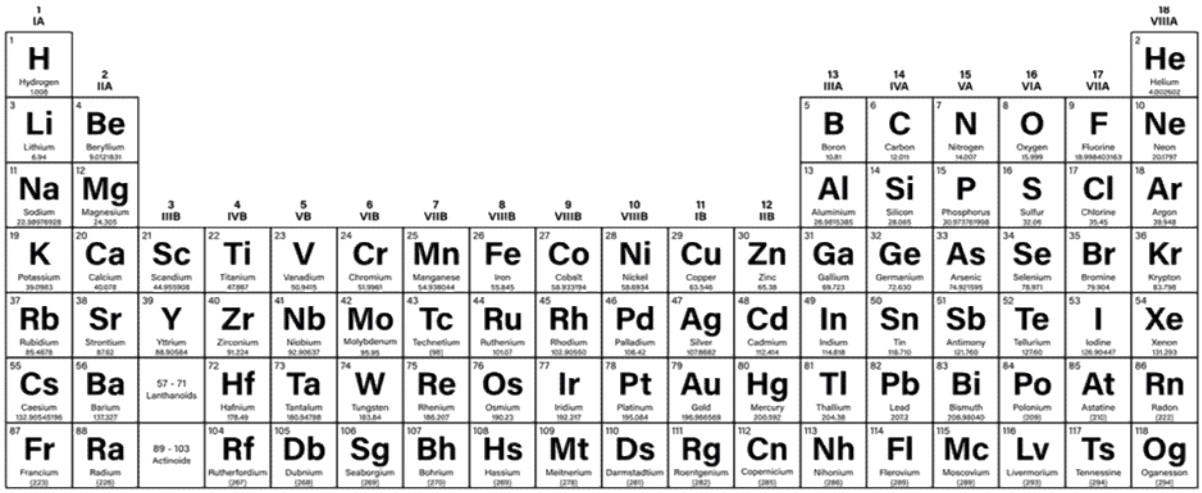
Instructions for CM2021 Mid-term Test

1. This test paper contains THIRTY (30) multiple-choice questions

2. Answer ALL questions (1 mark each)

3. Time allowed: 1 hour

4. Do not turn over the test paper until told to do so



1) Which of the following is not a coordination compound?

a. Transplatin
b. Haemoglobin
c. 1,2-Diaminoethane
d. Cobalt(III) ammines

2) The number of chlorine atoms acting as counteranions in the complex $[Co(en)_2(H_2)Cl]Cl_2$ is:

a. 1
b. 2
c. 3
d. None of these

3) Determine the number of structural isomers for the octahedral complex $[Co(NH_3)_5CO_3]Cl$.

a. 1
b. 2
c. 3
d. None of the above

4) Which of the following compound shows only linkage isomerism?

a. $[Co(NH_3)_5(SCN)](NO_2)$
b. $[Co(NO_2)(NH_3)_3]NO_2$
c. $[Co(NH_3)_6]Cl_2$
d. $[Co(NO_2)Br(NH_3)_4]Cl$

5) A metal complex has a general formula of $CoBr(NH_3)_5(SO_4)$. What is the identity of this metal complex if it produces white precipitate when reacted with barium chloride?

a. $CoBr(NH_3)_5(SO_4)$
b. $[CoBr(NH_3)_5]SO_4$
c. $[Co(NH_3)_5SO_4]Br$
d. $[CoBr(NH_3)_4SO_4].NH_3$

6) The name of the coordination compound $K_2[CoCl_4]$ is:

a. dipotassium tetrachlorocobaltate(II)
b. dipotassium tetrachlorocobalt(IV)
c. potassium tetrachlorocobaltate(II)
d. potassium tetrachlorocobalt(II)

7) The name of $[Fe(CN)_2((C_6H_5)_3P)_2]$ is:

a. dicyanobis(triphenylphosphine)iron(II)
b. dicyanodi(triphenylphosphine)iron(II)
c. dicyanobis(triphenylphosphine)iron(IV)
d. dicyanobis(tricyclohexylphosphine)iron(II)

8) The name of $[Cr(OH)_4]^-$ is:

a. tetraaquachromate(III)
b. tetrahydroxochromate(III)
c. tetraaquachromate(I)
d. tetrahydroxochromium(III)

9) Which of the following coordination complexes is most stable?

a. $[Ni(NH_3)_6]^{2+}$
b. $NiCl_2$
c. $[Ni(H_2O)_6]^{2+}$
d. $[Ni(en)_3]^{2+}$

10) An oxalato ligand has four oxygen atoms. It is a

a. monodentate ligand
b. bidentate ligand
c. tridentate ligand
d. tetradentate ligand

11) Which of the following gives the least number of isomers?

a. $[Co(NH_3)_4Cl_2]$
b. $[Ni(en)(NH_3)_4]^{2+}$
c. $[Ni(C_2O_4)(en)_2]$
d. $[Cr(SCN)_2(NH_3)_4]^{4+}$

12) $[Co(NH_3)_2(en)(NO_2)_2]Cl$ exhibits:

a. ionization, geometrical and optical isomerisms
b. linkage, geometrical and optical isomerisms
c. linkage, ionization and optical isomerisms
d. linkage, geometrical, optical and ionization isomerisms

13) The total number of possible isomers for this metal complex $[M(NO_2)(SCN)(NH_3)_4]$ is:

a. 3
b. 4
c. 8
d. 16

14) Which of the following compound shows optical isomerism?

a. $[Co(CN)_3(Cl)_3]^{3-}$
b. $[Cr(en)_3]^{3+}$
c. $[Zn(C_2O_4)_2]^{2-}$
d. $[Cu(NH_3)_4]^{2+}$

15) The number of geometrical isomers shown by the octahedral complex of the type $[M(A\text{-}B)_3]$ is

a. 1
b. 2
c. 4
d. None of the above

16) Which of the following metal atomic orbitals are considered in ligand field theory?

a. only d
b. p and d
c. s and d
d. s, p, and d

17) Which of the following can lead to stronger metal-ligand bond?

a. ligand-to-metal pi donation
b. metal-to-ligand pi donation
c. lower charge-to-radii ratio of metal
d. None of the above

18) What causes the d-orbitals to lose their degeneracy in the ligand field theory?

a. spherical field
b. point charges
c. mixing of metal atomic orbitals with ligand group orbitals
d. anisotropic crystal field

19) In $[Ti(CO)_6]^{4+}$, the Ti-C bond possesses:

a. π character only
b. σ character only
c. ionic character only
d. both π and σ character

20) In the complex ion ML_6^{n+}, M^{n+} has six d-electrons and L is high field ligand. According to crystal field theory, how many unpaired electrons are there in this complex ion?

a. 0
b. 1
c. 2
d. 4

21) Which of the orbital will be most stablized if two axial ligands (i.e., along the z-axis) are removed?

a. d_{xz}
b. d_{yz}
c. d_z^2
d. d_{xy}

22) What is the electronic arrangement of metal atom/ion in octahedral complex with d^4 configuration, if Δ_o > pairing energy

a. $t_{2g}^4 e_g^0$
b. $e_g^4 t_{2g}^0$
c. $t_{2g}^3 e_g^1$
d. $e_g^2 t_{2g}^2$

23) The ratio of CFSE for $[NiCl_6]^{4-}$ and $[NiCl_4]^{2-}$ will be nearly:

a. 3/2
b. 4/9
c. 9/4
d. 27/8

24) The d orbitals that are stabilized in a tetrahedral magnetic field are:

a. d_{xy} and d_{z2}
b. d_{x2-y2} and d_{z2}
c. d_{xy}, d_{xz} and d_{yz}
d. d_{z2} only

25) Which of the following is true when an octahedral metal complex undergoes axial compression?

a. d_{xy} orbital has the lowest energy
b. d_{x2-y2} orbital has the highest energy
c. Metal d-orbitals with lobes pointing towards the Z axis are stablized
d. All of the above

26) Which of the following on fluoro ligand is true?

a. σ-donor, π-acceptor
b. σ-donor, π-donor
c. σ-acceptor, π-acceptor
d. σ-acceptor, π-acceptor

27) In the spectrochemical series, which ligand is neither a pi acceptor nor a pi donor?

a. Cl^-
b. H_2O
c. NO_2
d. NH_3

28) Which effect/theory could account for the presence of two absorption peaks for a d^1 metal complex?

a. Jahn-Teller effect
b. Metal-to-ligand pi back bonding
c. Ligand-to-metal pi bonding
d. Mixing of metal atomic orbitals and LGO

29) In $[Cu(H_2O)_6]^{2+}$, the original aqua ligands can be replaced by ammine ligands. Predict the major product.

a. $[Cu(NH_3)_3(H_2O)_3]^{2+}$
b. $[Cu(NH_3)_4(H_2O)_2]^{2+}$
c. $[Cu(NH_3)_5(H_2O)]^{2+}$
d. $[Cu(NH_3)_6]^{2+}$

30) When $[Cr(CO)_5(PH_3)]$ is oxidized, what is the effect on the C-O bond?

a. Becomes stronger
b. Becomes longer
c. Becomes weaker
d. Both b and c

End of test paper.